Aerospace Manufacturing Core
Study Guide

"AMCSG-204"

Tools & Processes

Electricity

Electricity

Text from Google AI

Level Set

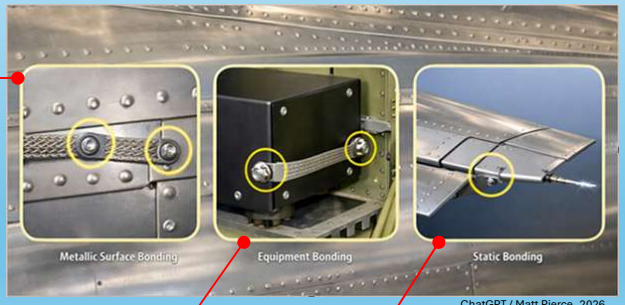
- **Electricity** = essential part of airplane
- Airplanes **use electricity** for flight controls, safety equipment, cabin amenities, avionics, hydraulic pumps, lighting, etc.
- Airplanes **use 40+ miles of wiring** to move electricity where it is needed
- **Bonding** and **Grounding** control flow of all electricity
  (wanted power and unwanted static)

Text from ChatGPT

Electrical Bonding

- **Electrical Bonding** = connecting all non-power carrying metallic objects in the airplane and airframe

- Three Types:
  - **Metallic Surface Bonding**
  - **Equipment Bonding**
  - **Static Bonding**



ChatGPT / Matt Pierce, 2026

- Metallic Surface bonding = Assemble metal-to-metal so electricity can flow out
- Equipment bonding = Attach equipment to metal airframe of body to prevent static electricity building / interfere radio and nav equip
- Static bonding = Attach isolated parts to the rest of the metallic bond

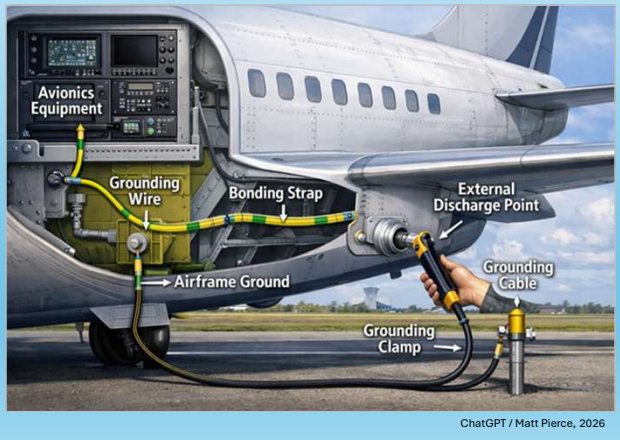
- Three Types Bonding Pic: ChatGPT: “make a picture of three type of bonding in an airplane: metallic surface bonding, equipment bonding, and static bonding”
  + “Combine these into one single composite image with three sections. Tune it specifically for maintenance training. Make examples in aluminum.”
  + “Making it one continuous fuselage scene with three highlighted zones”
  + “Adding subtle cues for inspection points (fasteners, straps, contact areas)“

Text from ChatGPT

Electrical Grounding

- **Electrical Grounding** = redirecting power spikes to airframe, then out external discharge points to atmosphere

- Grounding Prevents…
  - **Arcs** thru air between electrodes
  - **Static Electricity**; safe discharge
  - **Coupling** between circuits
  - **EMP** (only Faraday cage withstands)



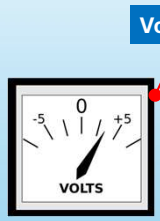
ChatGPT / Matt Pierce, 2026

- Grounding Pic: ChatGPT: “make a single picture of electrical grounding in airplane from a circuit to airframe then out external discharge point.“

Electrical Testing Equipment

- Electricity can be measured as
  - **Volts** (pressure)
  - **Amps** (flow rate)
  - **Ohms** (resistance/pipe size)
  - V = I x R
  - 1volt = 1amp x 1ohm

- There is a meter for each, or all in one:
  - Voltmeter
  - Ohmmeter
  - Ammeter
  - **Multimeter**



(PD) FreeSVG.org, 2017

(Ad) RS Products, 2026

(Ad) Grainger, 2026

(Ad) Walmart, 2026

Image Credits:

- **Voltmeter:** Google = (PD) - https://freesvg.org/analog-gauge
- **Ohmmeter:** RS Products = (Ad) - https://africa.rsdelivers.com/product/gossen-metrawatt/m630a/gossen-metrawatt-ohmmeter-10-m%CF%89/4375932
- **Ammeter:** Google = (Ad) - https://www.walmart.com/ip/Hyper-Tough-New-Digital-Clamp-Meter-TD35074B-6-4-oz-1-Count-1-25-in/410005314?wl13=2317&selectedSellerId=0&wmlspartner=wlpa
- **Multimeter:** For Sale / Public – https://static.grainger.com/rp/s/is/image/Grainger/F-15b-plus-01c__FHFE_v2?$adapimg$&hei=804&wid=804

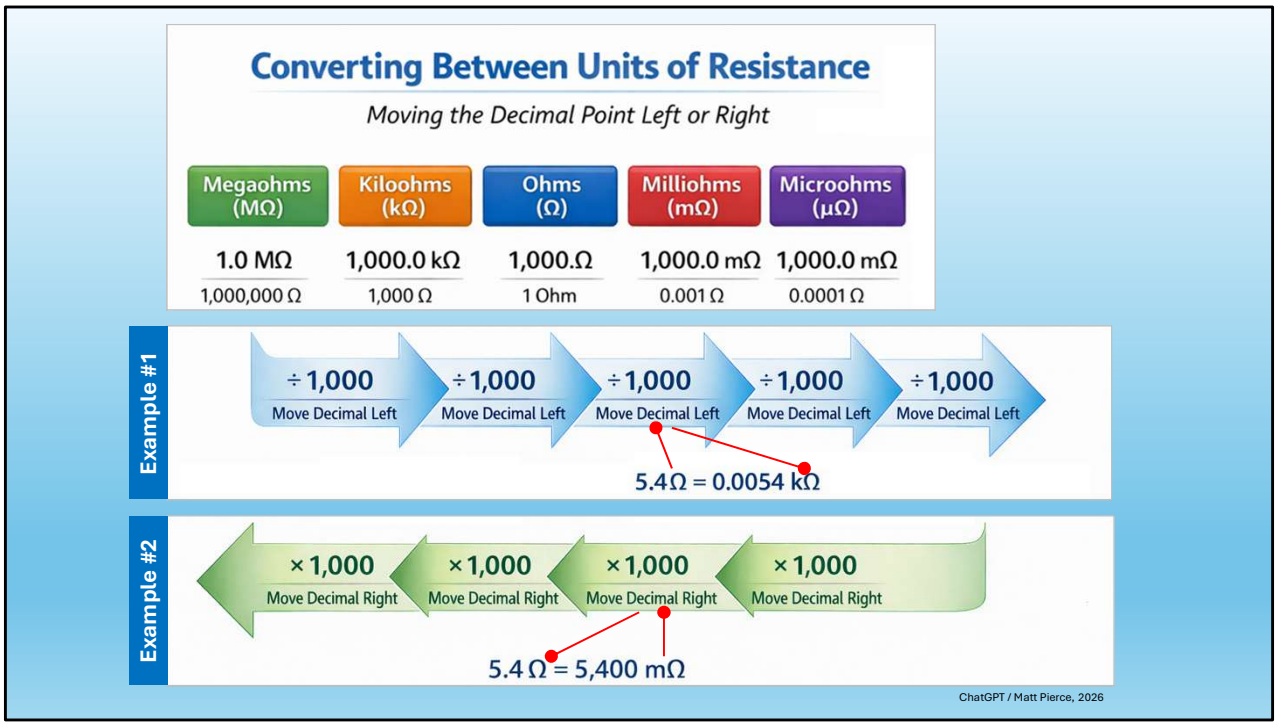


Image Credits:

- ChatGPT - make a diagram explaining how the decimals move left or right for megaohms to kiloohms to ohms to milliohms to microohms
  + Change the "+ 1,000" labels all to "÷ 1,000“
  + Add an example below the blue arrows (but above the green arrows) demonstrating how 5.4 ohms = 0.0054 kiloohms Add an example below the green arrows demonstrating how 5.4 ohms = 5,400 milliohms

Q&A

Practice Q&A's (Q1-Q8)

Q1: What is essential to an airplane for flight controls, pumps, equipment, lighting, etc.?

A: Electricity

Q2: Name of design that connects all non-power carrying metallic objects in an airplane to the airframe?

A: Bonding

Q3: Name of design that redirects spikes in power carrying equipment and lines to the airframe?

A: Grounding

Q4: How does electrical bonding work?

A: Wire straps connect to provide circuit between different metal surfaces, isolated metallic equipment, etc.

Q5: What four dangers does electrical grounding prevent?

A: 1. Arcs, 2. Static Electricity, 3. Coupling (between circuits), and 4. EMP

Q6: What is the formula representing the relationship between voltage, current, and resistance?

A: V = I x R; Voltage = Current x Resistance; 1 volt = 1 amp x 1 ohm

Q7: What is the measure of current?

A: Amperes

Q8: What is the measure of resistance?

A: Ohms

Practice Q&A's (Q9-Q10)

Q9: What is the measure of voltage?

A: Volts

Q10: What are the six size buckets for Ohms from left (largest) to right (smallest)?

A: Mega-ohms, Kilo-ohms, Ohms, Milli-ohms, Micro-ohms

Divide by 1,000 to shift right 1 (smaller units)

Multiple by 1,000 to shift left 1 (larger units)

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg